

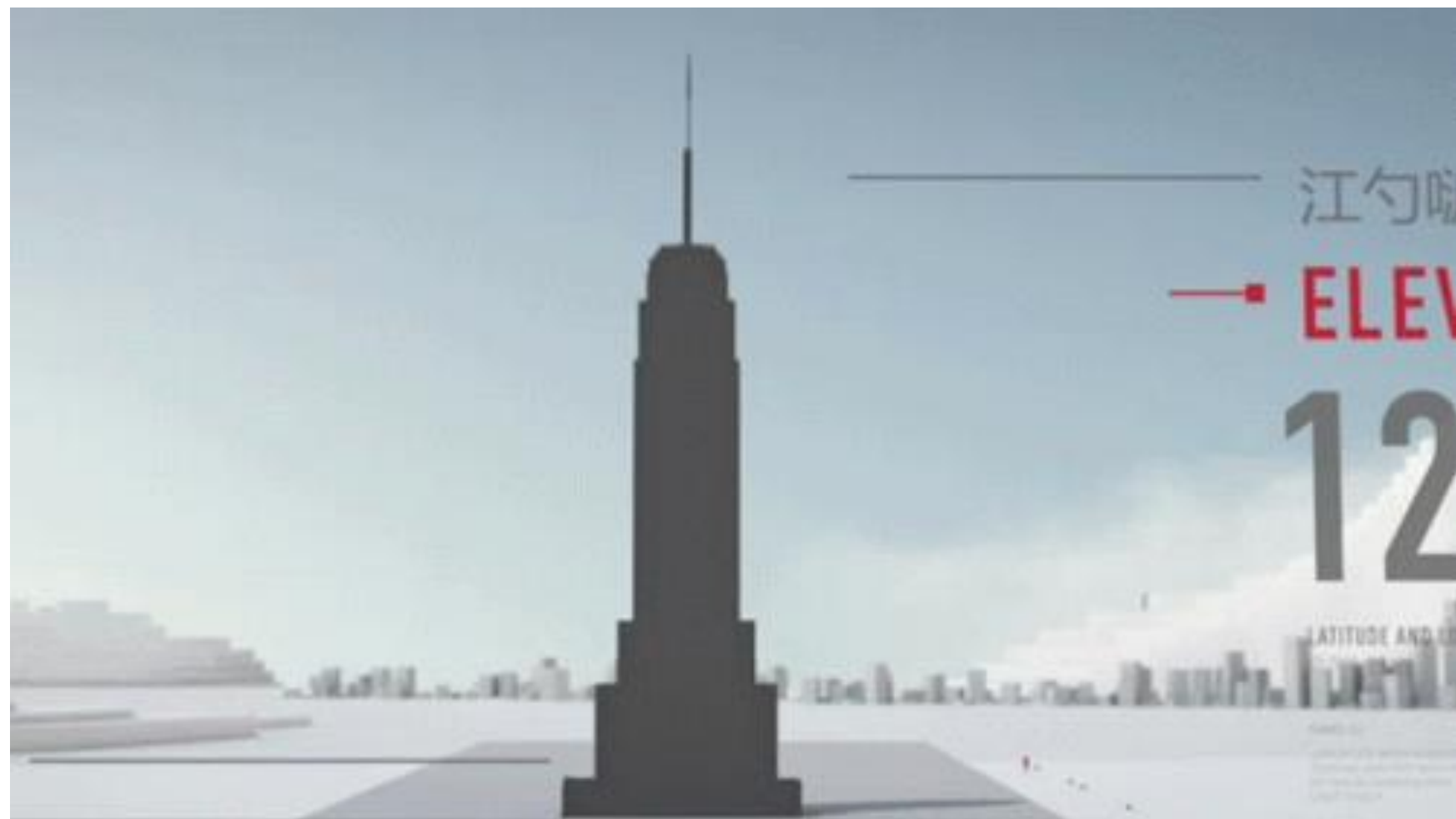
SHTeam Labs

Hexagons #1 - 25

WEEK ONE

Hexagon 11 & 12

**Paper Tower
& Group Reflection**



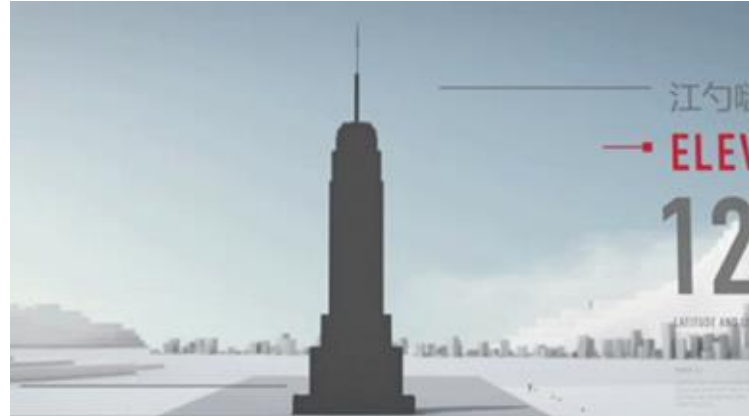
Seating For Today

Grab a chair or stool,

max 5 per table,

you can sit at the hot glue stations as well
(though no hot glue today)

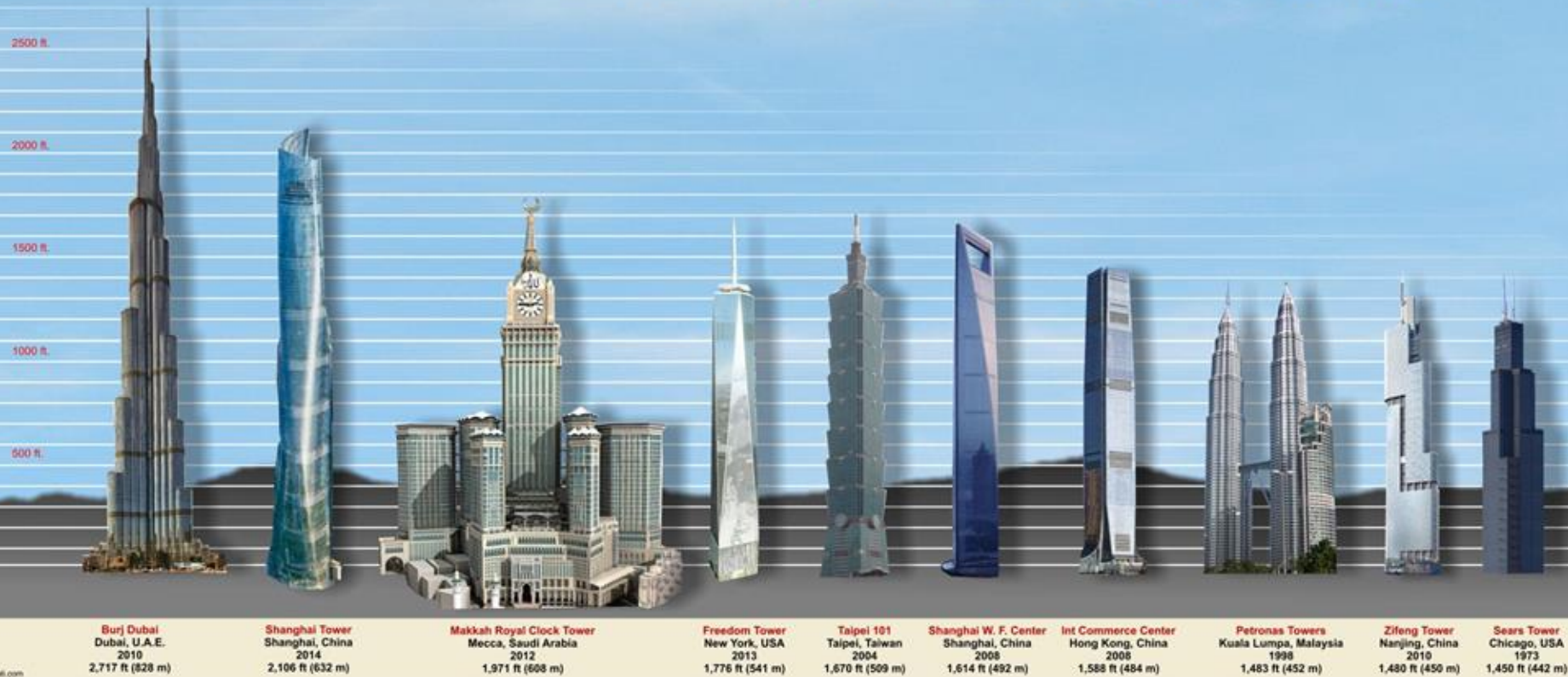
Do people
make things
by
themselves or
with others?



Solo Structure
Single Sheet of Paper

Who can make the
tallest? You have 3 mins

THE TOP TEN WORLD'S TALLEST BUILDINGS



Paper Tower Challenge #1

11

- **Your goal: Build the tallest paper tower**
- **Our goal: Teamwork Test!**
 - Can you compete AND help each other?
 - When can we use more tools?
 - More Tools are: hot, sharp, tiny, messy, or valuable



Skill Test

TinkerCAD

OctoStudio

Micro:bit MakeCode

Adobe Express

Identify Your Business

West of 19th Street

South of Lincoln Way

North of Sloat Boulevard

Probably not a big chain.

Tutorial Research!

Using online tutorials to complete complex projects is a valuable skill.

Activities

1. Begin the lesson by showing students a short video of famous towers around the world, like the Eiffel Tower or the Burj Khalifa, and ask them what makes these towers stand tall and strong. This will spark curiosity about the principles of building structures.
2. Provide each student with a single sheet of paper and challenge them to make it stand as tall as possible without any additional materials. This quick activity will encourage creative thinking and set the stage for the paper tower challenge.
3. Organize a storytelling session where you narrate a fun story about a group of animals trying to build the tallest treehouse. Encourage students to discuss what strategies the animals might use to make their treehouse stable and tall, linking these ideas to the upcoming paper tower challenge.

Student Goal: Build the tallest free standing paper tower, your class will be competing against the other classes

Teacher Goal: How well does the **class work together?**

Requirements:

- Must be free standing: Cannot lean it against anything, Cannot hold it up
- Must stand on its own for 10 seconds
- Only paper, no other materials
- After activity – leave the makerspace better than you found it

Constraints:

- No tools to aid your building
- Cannot anchor base, this includes hands, taping, or gluing
- 10 mins to build

Reflection

How were you excellent to another person?

What could be improved next time?

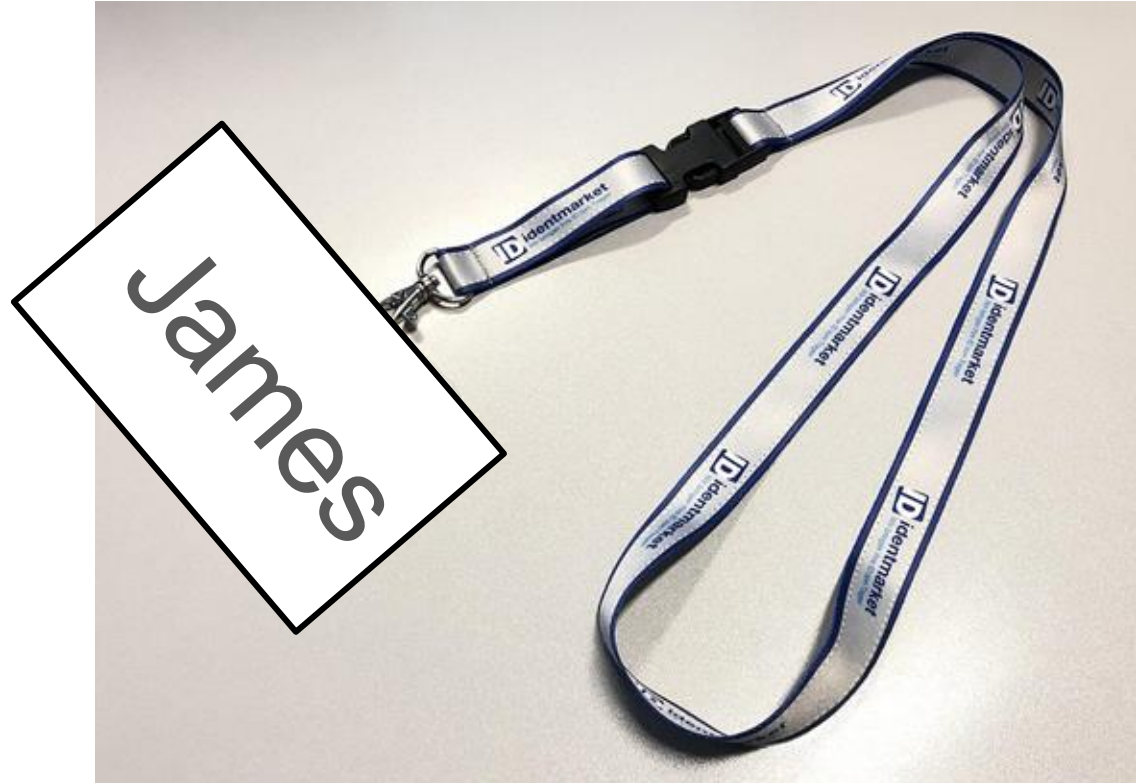
What went well?

Nametags

When your tower
is complete,
make a name
tag!

Big clear letters
Black maker

Throw away the
old name



Fast Feedback Protocol

If we have time, we will do Fast Feedback lines! An easy topic, like summer activities or upcoming school hopes.

If you are working on your name tag, observe us after you finish.

HEXAGON #11

Skill

Engineering Skills: Do It Together

Activity

Paper Tower Challenge #1: How well do we work together?

Goal: Build the tallest free standing paper tower

Materials:

- Paper
- Lots of clear floor or table space
- Meterstick
- Timer

Requirements:

- Must be free standing
 - Cannot lean it against anything
 - Cannot hold it up
 - Must stand on its own for 20 seconds
- Only paper, no other materials
- After activity - leave it better than you found it

Constraints:

- No tools to aid your building
- Cannot anchor base, this includes taping or gluing
- 15 mins to build

REFLECTION

Hexagon #

Date

What worked well?

What can you do better next time?

How were you excellent to yourself and each other?

HEXAGON #12

Skill

Engineering Skills: Self-Reflection

Activity

Engineering Reflection and Feedback Session

Materials:

- Poster paper, one per team
- Sticky notes
- Markers

Words to know:

- Reflection - thinking about your thoughts, actions, feelings, and choices good or bad. When you reflect you are like a detective on the things you did, how you did them, and why you did them so that you can progress.
- Feedback - is getting or giving advice. It should be helpful, useful, and kind. Feedback should be about the work and not the person. Sometimes it may be hard to hear, but the purpose of feedback is so we can improve.

Do:

1. Stay in your groups
2. Follow instructions from the teacher
3. Remember this is about progress, so don't forget to be excellent to one another
4. Present your posters

REFLECTION

Hexagon #

Date

I learned...

I want to know more about...

Notes / Feedback



What makes a product
useful?

WEEK TWO

Hexagon 15

Paper Headwear
Challenge

Paper Headwear Challenge

- Your goal: Engineer a stylish paper headwear for yourself
 - Secret goal: Can you help other students, and receive help?
-
- Game Jams: Engineering comes in many different forms
 - Activity reflections - Circle
 - What kind of team work is needed to manufacture a hat?
 - What type of skills are needed?
 - Problem solving and creativity are related













Individual Goal: Engineer and build a stylish headwear, for yourself! Make it useful for 3 different reasons.

Group Goal: Help fellow classmates with techniques

Requirements:

- Steady structure - it stays on your head when you are dancing. No hands!
- Material - only paper, tape, and coloring with crayons
- Share in a gallery walk

Constraints:

- No more than 3 sheets of paper
- Time: 12 mins

Useful Things

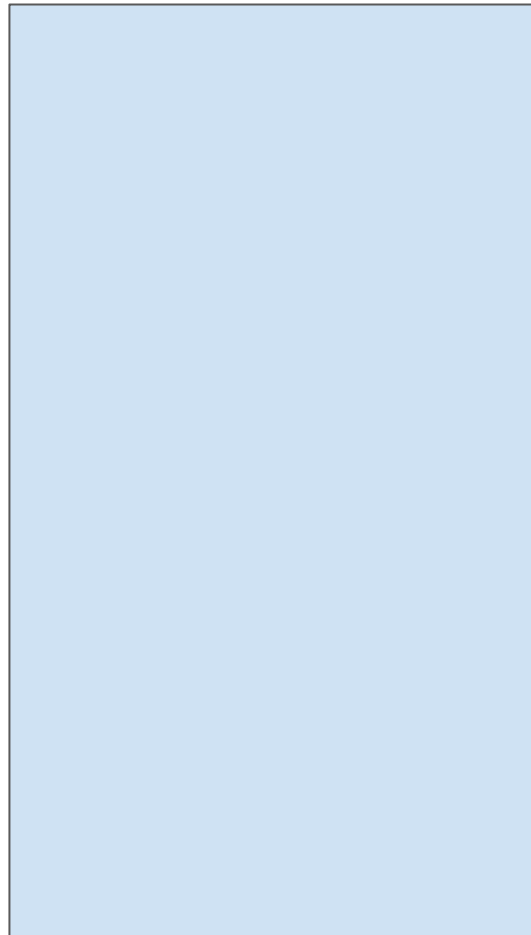
Can it stay on as you move around? Dance to test it

Do you like how it looks?

Does it tell anyone about you?

Does it fit well? Can another person also wear it?

How can it be useful? How can I make it even more useful?



Reflection

Would you craft the hat differently if you were making it for one of your teachers? Why?

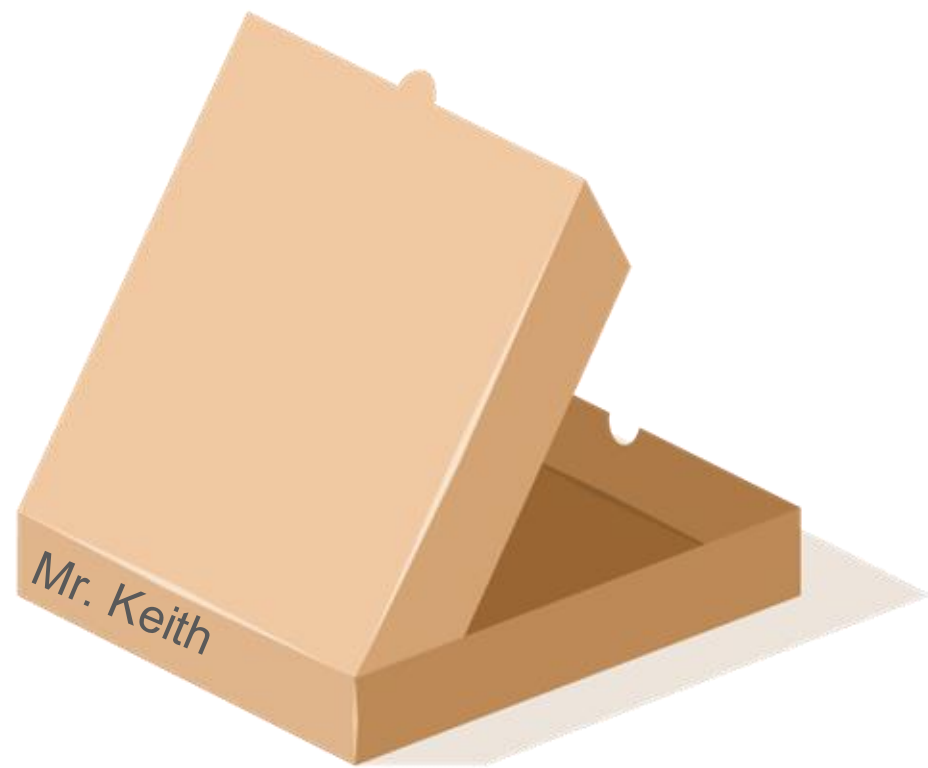
How can an engineer design for someone else? What could they think about, ask, or do?

What is your favorite tool and
why?

WEEK THREE

Hexagon 6, 7, x

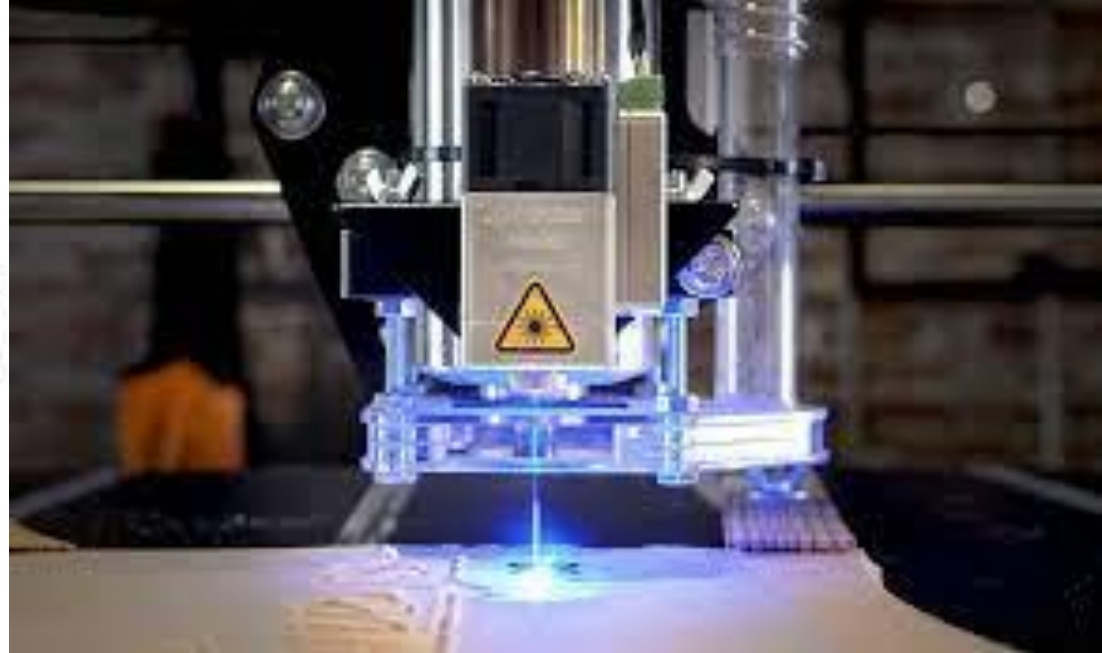
- Project Box
- Cardboard Cutter Safety + Use





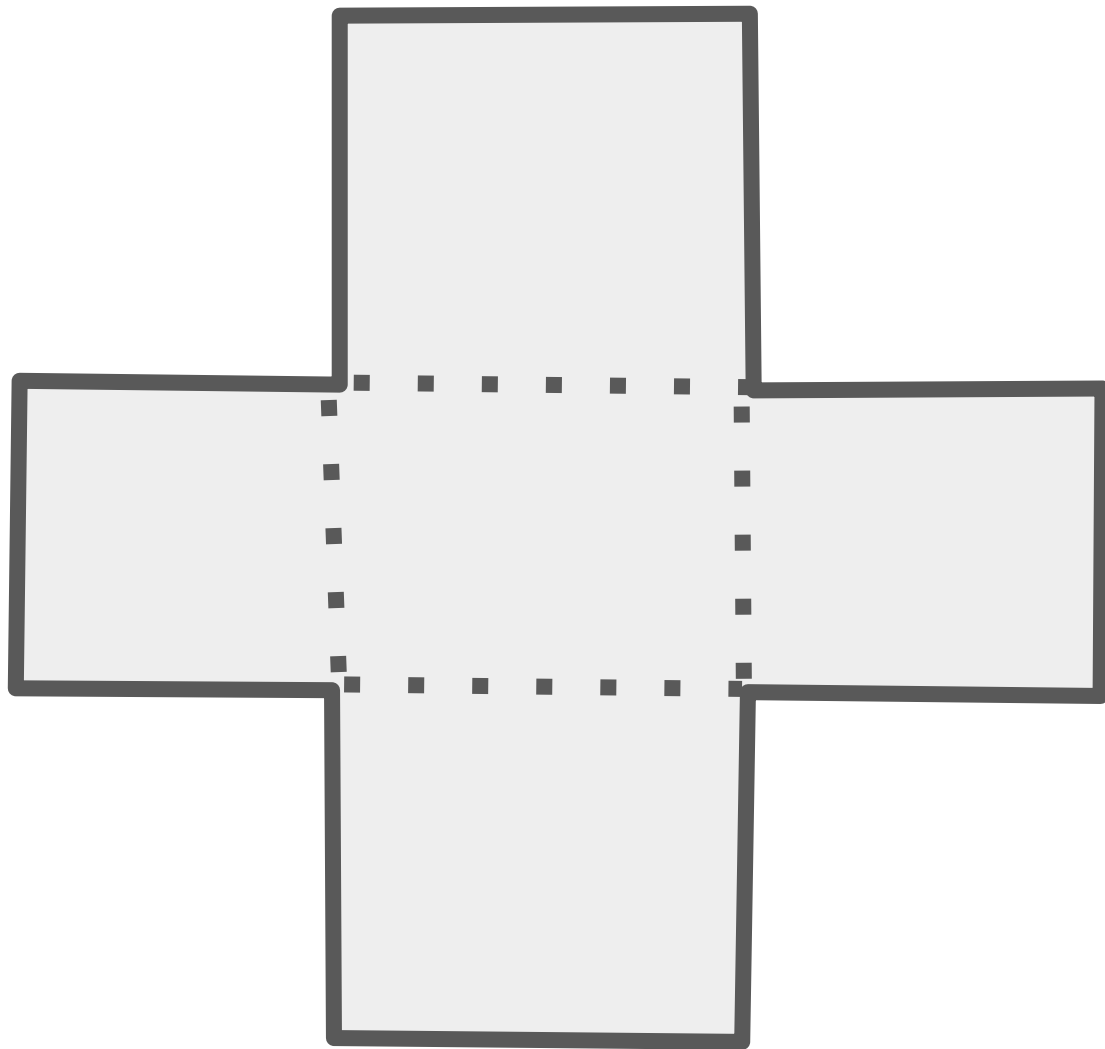
**Technology Toolbox
for Educators**





WEEK FOUR

Why do we measure
things?



CARDBOARD ATTACHMENTS



FLANGE

L-BRACE



SLOT



CUT

+



CUT

=



INSERT

TABS



TAB

+



SLOT

=



BRASS FASTENERS



HEY-IT MOVES!

HOLE PUNCH
+ WIRE



FOOT

ATTACHMENT Techniques



Flange



Gusset $\Rightarrow$ Δ supports



L-brace

Slot



cut

+



cut

=



insert

Tabs



tab

+



slot



Brass Fastener



HEY-IT MOVES!

Hole Punch
+ Paperclips



skewer
(cardboard moves)



↑
Flange



Gusset $\rightarrow$ Δ supports



L-brace

Slot



+



=

cut



insert

cut



Tabs



+



tab



=

slot

Brass Fastener



Hole Punch
&
Pip cleaners



Hexagon #10

- Construct a cube
- Cutter & Adhesive
- Our first client

- Why do humans make? Because it is useful - here's our first client!
- Ms. Kylie seedling, Kindergarten crayon holders, Ms. Carrie - LED or microbit sorter
- Goal Prototype something useful for one of our clients
- Requirements:
- Constraints:

Activity Choices

- Big Lego
- Magnetic Tiles
- Big Geometry
- Weaving
- Knotting

Last class we measured, we cut,
we made things.

What makes something useful?

Hexagon #10

Your goal: Prototype a useful box for a client

Makerspace goal: See how you combine your skills and see how you practice safety.

1. Pick a client.
2. In Hexagon #10, in the notes section brainstorm what you think your client needs and wants.
3. Prototype - Each person needs to make their own box, but we will work in tool groups of no more than 3.
 - Take a template if you need one.
 - Safety reminders
 - Hot Glue Gun Station - ONLY 4 at a time - LINE UP - DO NOT DO THE WORK AT THE STATION
 - Use the cardboard from the edges towards the center

Hexagon #16

- Prototype Solution to Problem
- Intro Design Thinking
- Empathy Focus

“Empathy is being able to put yourself in someone else’s shoe. It means you understand how that person feels and understand the choices they make.”

Why is empathy important when designing for a client?

(Why is it important to understand your client?)



OUT TO LUNCH

IF T-REX
HAD MODERN
TECHNOLOGY!



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the-cartoonist.com

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Client #1: Ulloa and Sunset Kindergarteners and Crayons



Problem: Kindergarteners would like to have more crayon holders at their table.

Requirements:

- Crayons do not fall out from the bottom.
- Easy to pull out and put back crayons.
- The holder should look nice.

Client #2: Ms. Coral Bells and Gardening



Problem: We need pots for seed starting with 2nd - 4th graders.

Requirements:

- Holds the soil and seed.
- Can transport without the pots falling apart.
- It will start one plant per pot.

Client #3: Sunset Makerspace and Hot Glue Gun Users



Problem: Easier to have glue sticks for available at the hot glue gun station.

Requirements:

- Holds 4 - 8 glue sticks.
- It is sturdy and will not fall over.
- Portable.
- Easy to know it's for hot glue sticks.
- Safety considerations.

Client List

1. Ulloa and Sunset grade K crayons
2. Gardening Boxes
3. Hot Glue Stick Boxes
4. Desk supplies case
5. Micro:bit case

Hexagon #16

Your goal: Prototype a solution to a problem for a client of your choice

Carrie goal: See how you practice empathy, think before doing, and see how you practice safety.

Requirements:

- Must be useful - it must solve your client's problem
- Must be beautiful
- Must get approval for your materials

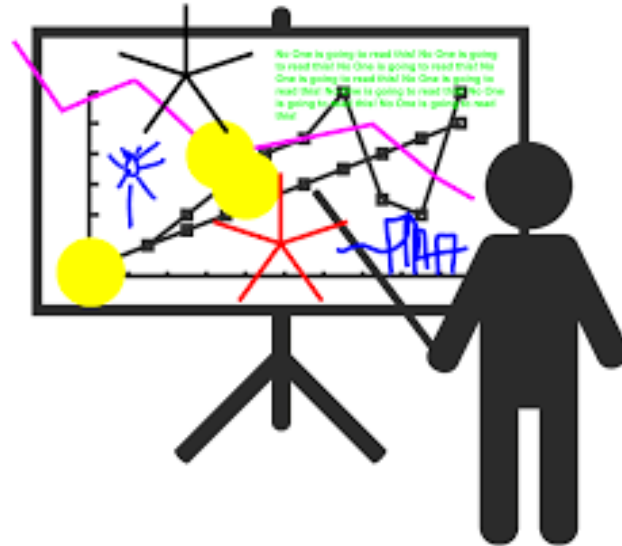
Constraints:

- Only can use tools in the tool caddy
- No more than 7 items from the table. There are items that say only one.
- Time: 10 - 15 mins

Hexagon #14

- Cardboard Joints & Attachments
- Presentation “Final Product”

What makes a good presentation?



Hexagon #14

Your goal: Make a poster for 1st graders to show them how to connect cardboard and paper (Not a prototype! A final product!)

Carrie goal: See how you practice empathy, creativity, and teamwork.

Requirements:

- The poster must be organized and clearly labeled.
- The poster must look nice for our clients (1st graders).
- The attachments must stay on the poster. It will not fall off if we hang it up.
- Names on the bottom right corner.

Constraints:

- Teams of three or less - one tool caddy per team.
- Only materials on the materials table.
- Time: 10 - 15 mins

CARDBOARD ATTACHMENTS



FLANGE

L-BRACE



SLOT



CUT

+



CUT

=



INSERT

TABS



TAB

+



SLOT

=



BRASS FASTENERS



HEY-IT MOVES!

HOLE PUNCH
+ WIRE



FOOT



ATTACHMENT Techniques



Flange



Gusset $\Rightarrow$ Δ supports



L-brace

Slot



cut

+



cut

=



insert

Tabs



tab

+



slot



Brass Fastener



HEY-IT MOVES!

Hole Punch
& Pencil sharpeners



skewer (cardboard moves)

3D Paper Techniques



Artemis Launches Astronauts

Astronaut Talk: <https://youtu.be/6GSqfURNOa4?si=tcoP653aEVd4TX0R&t=187>

The Launch: https://www.youtube.com/shorts/vMGuObY8_sw

Extended Launch 360: <https://www.youtube.com/watch?v=9LFBFRH51Jo>

Flight Path: <https://www.cbc.ca/kidsnews/post/latest-updates-artemis-ii-moon-mission>

Design Challenge!

Goal: Make a holder for an astronaut item of your choice - food, waterbottle, plushie, etc.

Requirements:

- It is secured to the astronaut's body.
- It is easy to put the item on and remove it from the holder.
- It is easy to put it on and take it off the body.

Constraints:

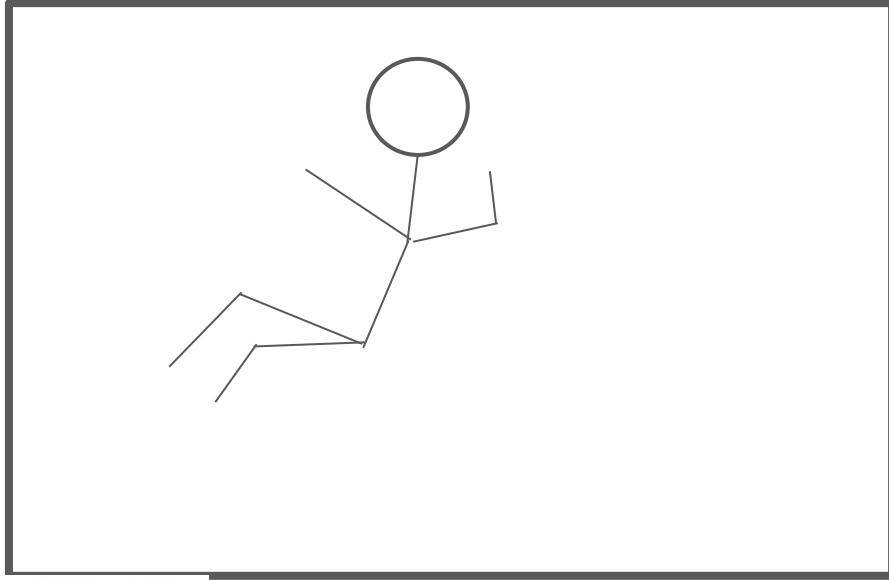
- No gravity!!
- 10 mins to make.

Design for Space: Item Holder for Astronauts



Item _____

Body Part _____



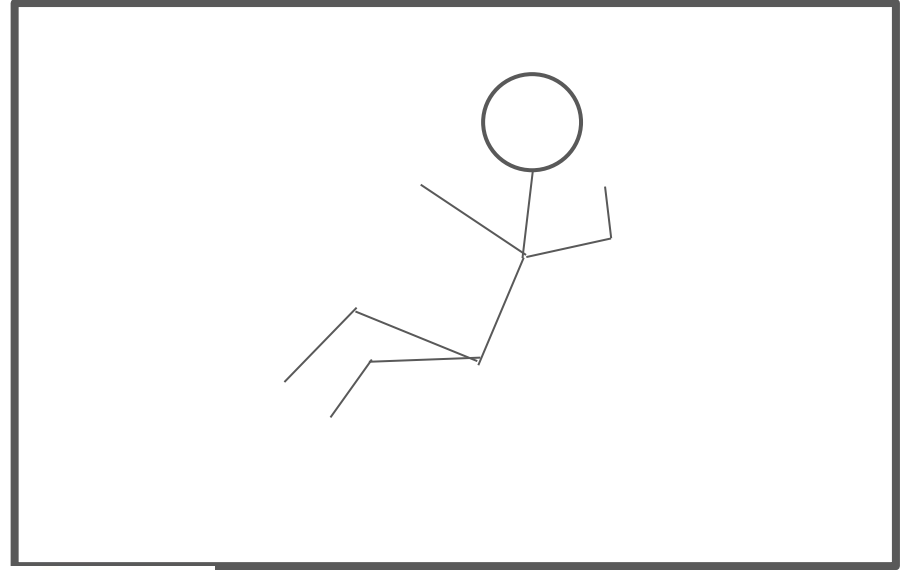
of Cardboard: 1



of Paper: 1 2 3

Item _____

Body Part _____



of Cardboard: 1



of Paper: 1 2 3

Hexagon #26

- Intro to microbits
- Use code to control LEDs and input

Hexagon #26

- Intro to microbits
- Use code to control LEDs and input

Besides computers and phones, what are three things in your house that have a computer chip inside?







Hexagon #26 & #2

Your goal: With a microbit -

- Output: Control LEDs with code - download a flashing heart to the microbit
- Input: Use the shake block code

Carrie goal:

- Engineering Skill: See how resourceful you are when something does work the way you want
- Maker Mindset: See how you help one another EVEN if it's not your friend.

Do:

- Know how to get on code editor
- Know how to use tutorial
- Know how to start a new project - challenge: use shake block

makecode.microbit.org

Maker Challenge!

Have the microbit show your name when you shake it! Show your last name when you tilt left.

Hexagon #26

- Intro to microbits
- Counting and Intro to Accelerometer

Why do humans count things?
What if we didn't?

What are some creative ways
people communicate with
each other?



Fast Prototype

Requirements:

Pizza Box closes at the end of each class

Integrates a microbit into the game

This will be the first prototype, maybe just 4 classes. we will have another final project

Hexagon #27 & #5

Your goal: With a microbit -

- Output: Show how many steps/jumps you have taken on the microbit
- Input: Use an **accelerometer** to sense your steps/jumps taken

Carrie goal:

- Coding Skill: See how you use **logic** to tell the microbit what to do
- Science Skill: Practice **collecting data** and **making the data useful** for you

Do:

How we are going to do this:

- 1) Code it - 6 mins
- 2) Attach it (slip it in your laces or swing your arms) - 2 mins
- 3) Collect the data (Walk to the exit and back quietly) - 2 mins
- 4) Record the data - 2 mins
- 5) Team Meeting (Wonderings, Shares, Is this data useful?) - 2 mins
- 6) Hack the Code

How we are going to do this:

- 1) Code it - 6 mins
- 2) Attach it (hold it in your hand or put it in your pocket) - 2 mins
- 3) Collect the data (jump! Jump safely please.) - 2 mins
- 4) Record the data - 2 mins
- 5) Team Meeting (Wonderings, Shares, Is this data useful?) - 2 mins
- 6) Hack the Code

Watch! Microbit:
Accelerometer

<https://youtu.be/UT35ODxv>
[mS0](#)

Makerspace Google Class Codes

3rd Grade: dxbd4rd

4th Grade: 2j6bpey

5th Grade: niuvjx5

We did the skill
interviews instead of
the waterbottle
-Sunset 2024

Hexagon #19

- Water Bottle for a superhero
- Function vs Aesthetics

Looks vs. Use

Is one more important than the other?

Hexagon #19

Your goal: Prototype a water bottle holder suitable for your client

Carrie goal: See how you think about the differences between wants and needs, observe how plan before doing, and see how you practice safety.

Requirements:

- The water bottle must stay on your client's body while they are doing heroic things.
- The water bottle must not put them in danger.
- The water bottle must be easy to drink from.
- Must get approval for your materials

Constraints:

- Only can use tools in the tool caddy
- No more than 7 items from the table.
- Time: 10 - 15 mins





Skill

Design Skills: Learning Station

Activity

What is Design? Use vs. Looks

Goal: Know the difference between functional and aesthetic design by designing a nice home for your client

Words to know:

- Functional - how something is used
- Aesthetic - how something looks and makes you feel

Materials:

- Clients to design for
- Pencil and color pencils
- (Optional) Device for research
- (Optional) Dotted or grid paper

Do:

1. Pick one of the clients at the Learning Station.
2. Follow the guide for your design.
3. Reflect.

HEXAGON #19

Skill

Design Skills: Learning Station



Design Guide

Design Guide: What is Design? Use vs. Looks

Client Name:

Client Needs:

Client Wants (Likes):

Sketch a design with explanations labels: (Need more paper? Get it!)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
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Hexagon #30

- Electricity, Circuits
- Make own circuit
- what makes a game a game?

- AMA
- What is electricity? What do we use it for? - Round robin
- What is electricity? Video
- Controlling electricity - Activity
- What makes a game a game?
Separate poster papers per class

WELCOME TO THE MAKERSPACE

repeat every day

When you walk inside

be safe

be respectful

be responsible

Think about three things you use
that needs electricity.

Hexagon #30

Your goal: Control electricity! Make your first circuit!

Carrie goal: See how you use materials appropriately like an engineer. See how you try to figure things out for yourself like a coder. See how you help each other out like a manager.

Requirements:

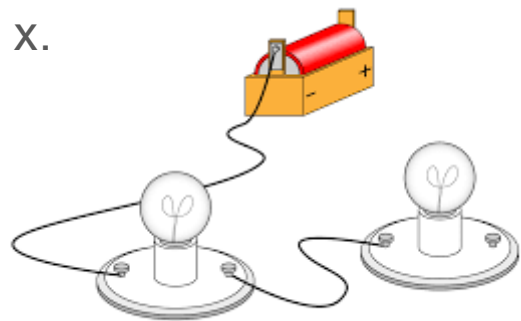
- Turn on your LED in two ways
- Be able to explain your circuit and which way the electricity is flowing.
- Add arrows to your circuit to show which way the electricity is flowing.
- Know your components: LED, copper tape, battery

Constraints:

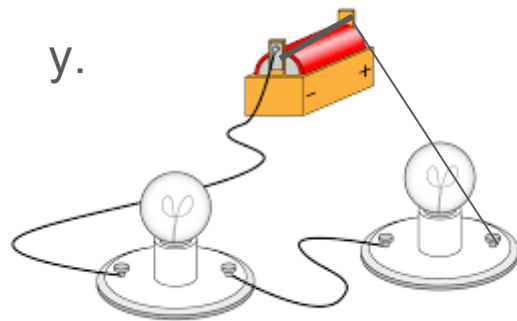
- Only can use materials in the materials caddy
- You can only take the copper circuit and LED (if you want it). LEAVE EVERYTHING ELSE
- Time: 10 - 15 mins



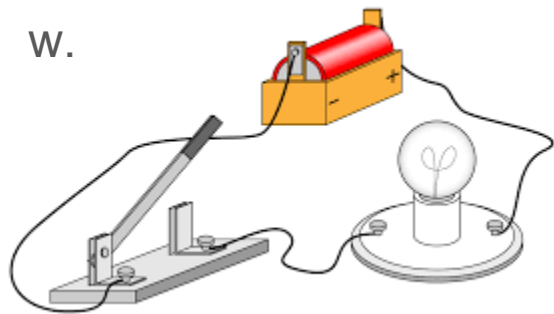
x.



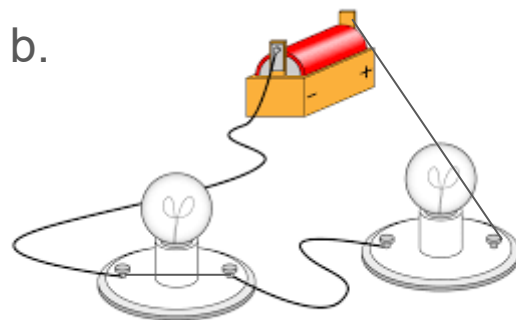
y.



w.



b.



What makes a game a game?
(What must a game have?)

Hexagon #31

- Conductive vs. Insulator
- Make own conductivity tester

Hexagon #31

Your goal: Be able to explain what conductor and insulators are AND give examples.

Carrie goal: See how you use materials appropriately like an engineer. See how you try to figure things out for yourself like a coder. See how you help each other out like a leader.

Requirements:

- Use the microbit as a tool - we will code a conductivity tester.
- Fill out the Data Sheet with your partner on the table.

Constraints:

- Leave it better than you found it for the next class.
- Time: 10 - 15 mins

Hexagon

- Space Station Modules
- LED blinker buttons

Short Teacher Summary

Students explore the idea that space stations can grow by adding modules.

- After viewing media and discussing the ISS & future stations,
- and the usefulness of LEDs in multilingual, high-tech environments,

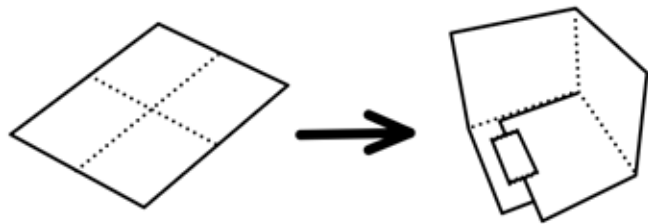
Each student builds a paper module that stands upright and serves a function within the station.

Students then add a simple LED-and-battery component that supports a specific activity or signal in their room. Finished modules are placed together on a black-fabric table to create a class-built space station.



CARA DELEVINGNE

A. Square Corner

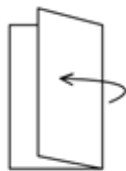


1.



1 sheet of paper. It can be a square or rectangle

2.



Fold paper in half, **short edge to short edge**

3.



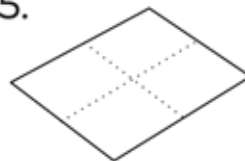
Unfold

4.



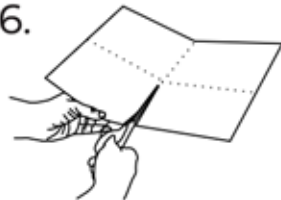
Fold paper in half, **long edge to long edge**

5.



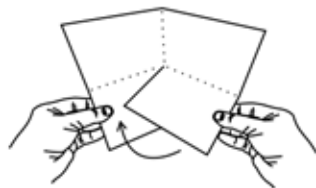
Unfold

6.



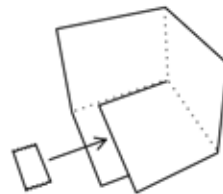
Cut on crease from **long edge** to center

7.



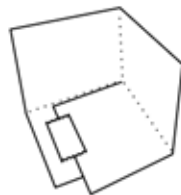
Overlap the flaps

8.



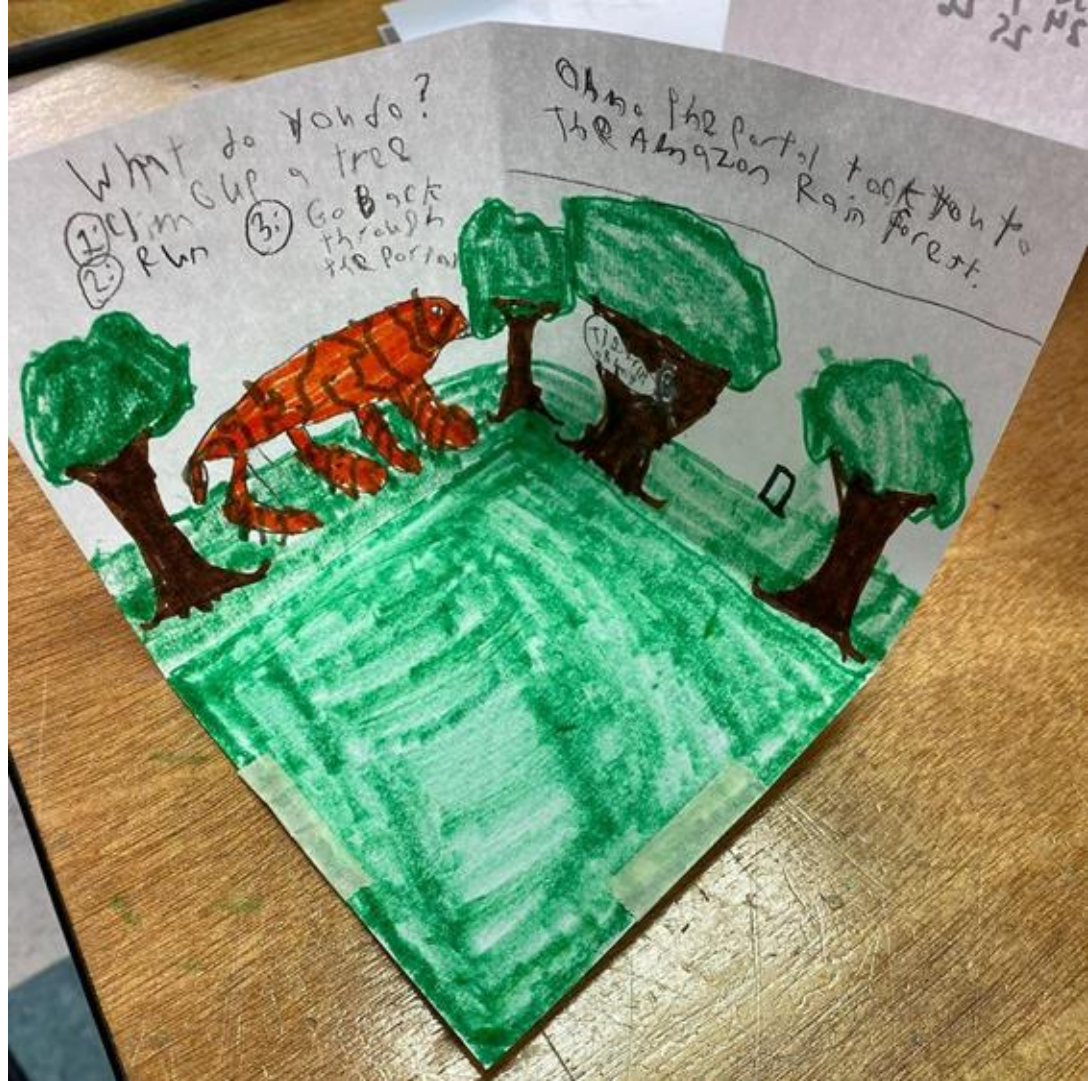
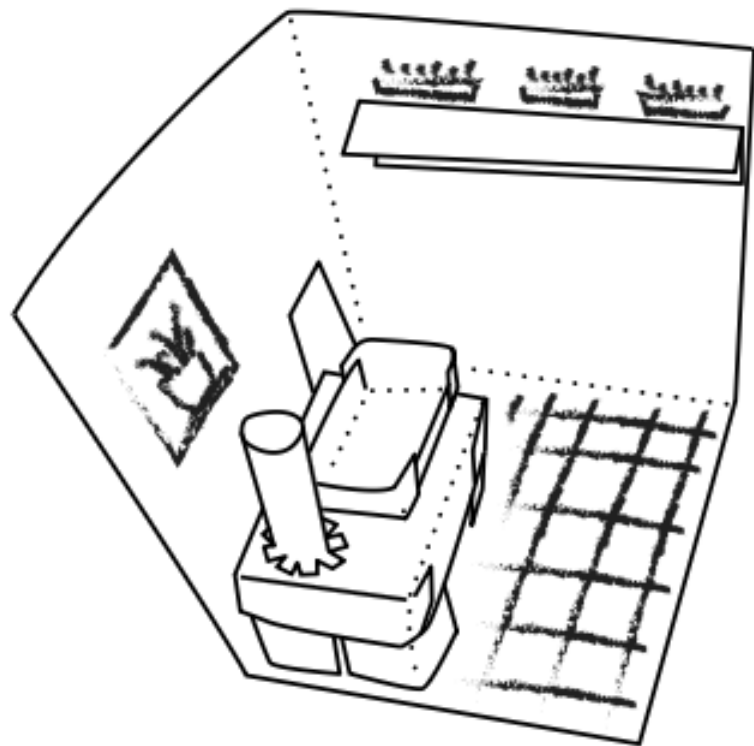
Tape in place

9.

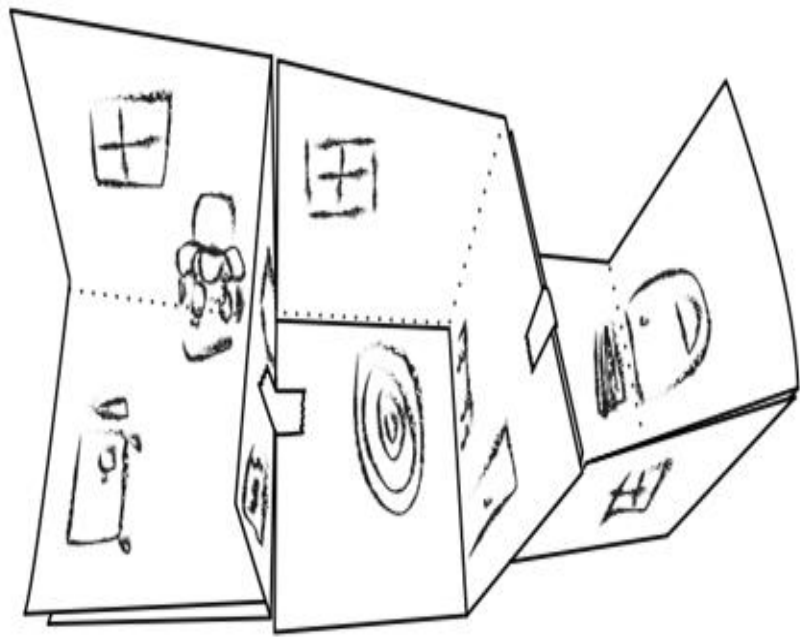


Finished

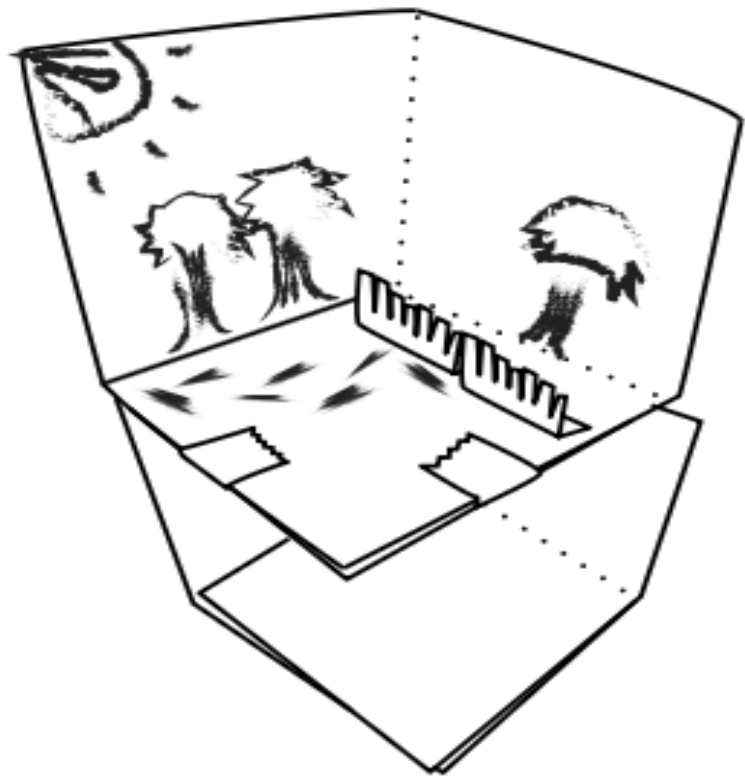
Single



Side-by-Side



Stacked

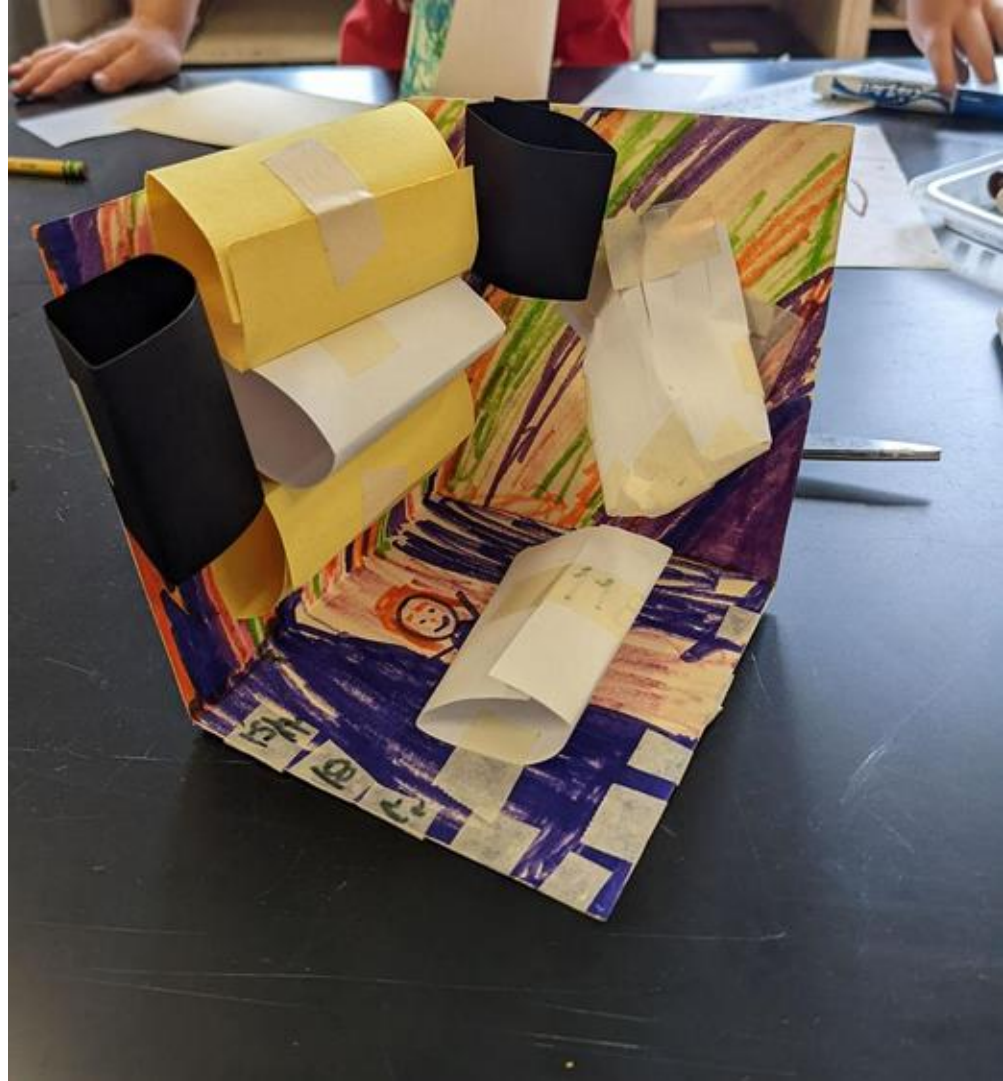


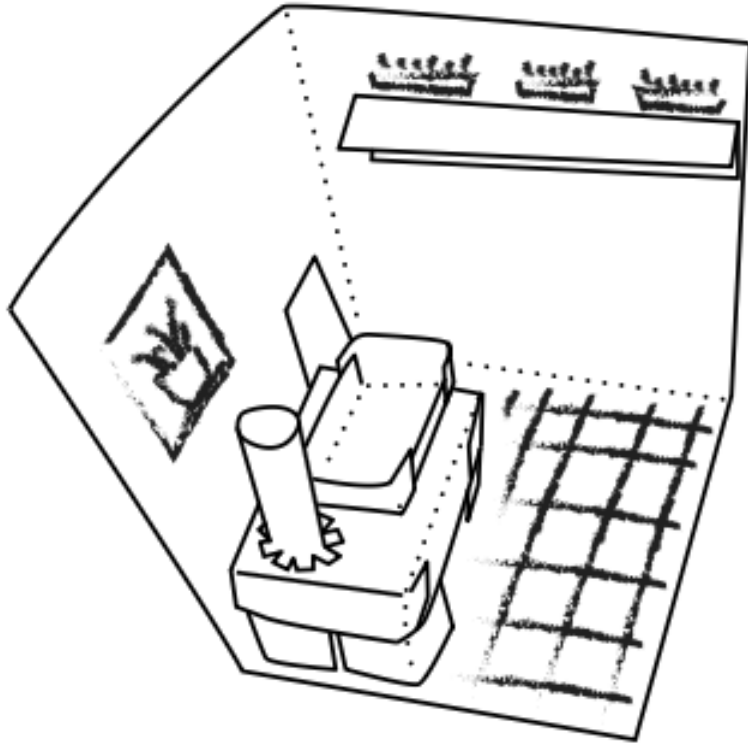
With Paper Furniture

What features should your Space Station Module have?

What activities would you be doing in zero gravity?

What if you are there all day, 24 hours, for several days?



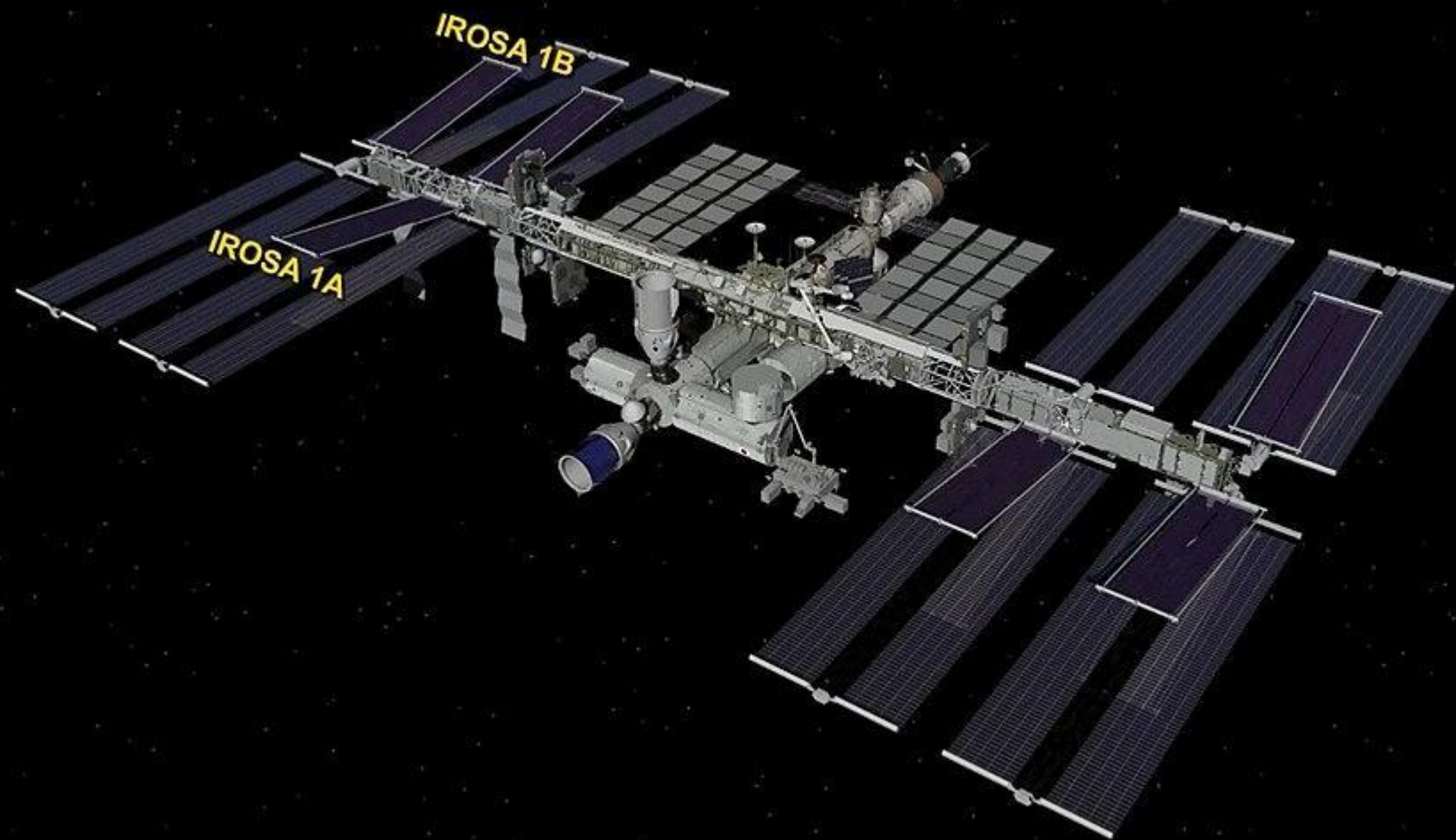


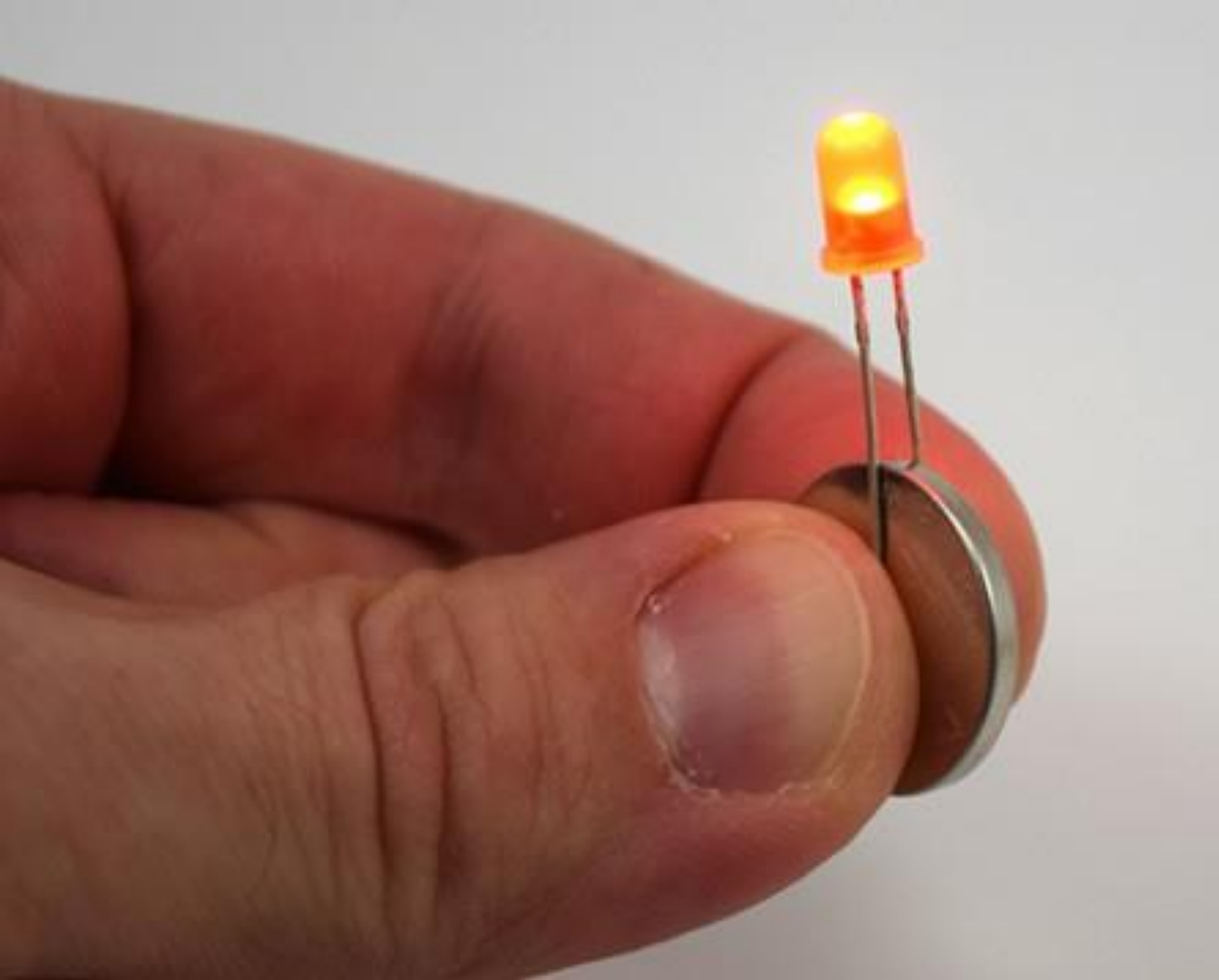
Paper Module Requirements

Dock: A place to connect to another module

Life support: food, water, and a place to sleep

Mission: at least one activity can be done there





LED

Pick a color, and
add your LED!

Can you attach it
with a tape loop?

Can you make it
turn on when you
press it?

Hexagon #32

- Circuits - parallel vs series
- Motor and LEDs
- What makes a good game

Hexagon #32

Your goal: Be able to explain how parallel and series circuits are different

Carrie goal: See how you use materials appropriately like an engineer. See how you try to figure things out for yourself like a coder. See how you help each other out like a leader.

Requirements:

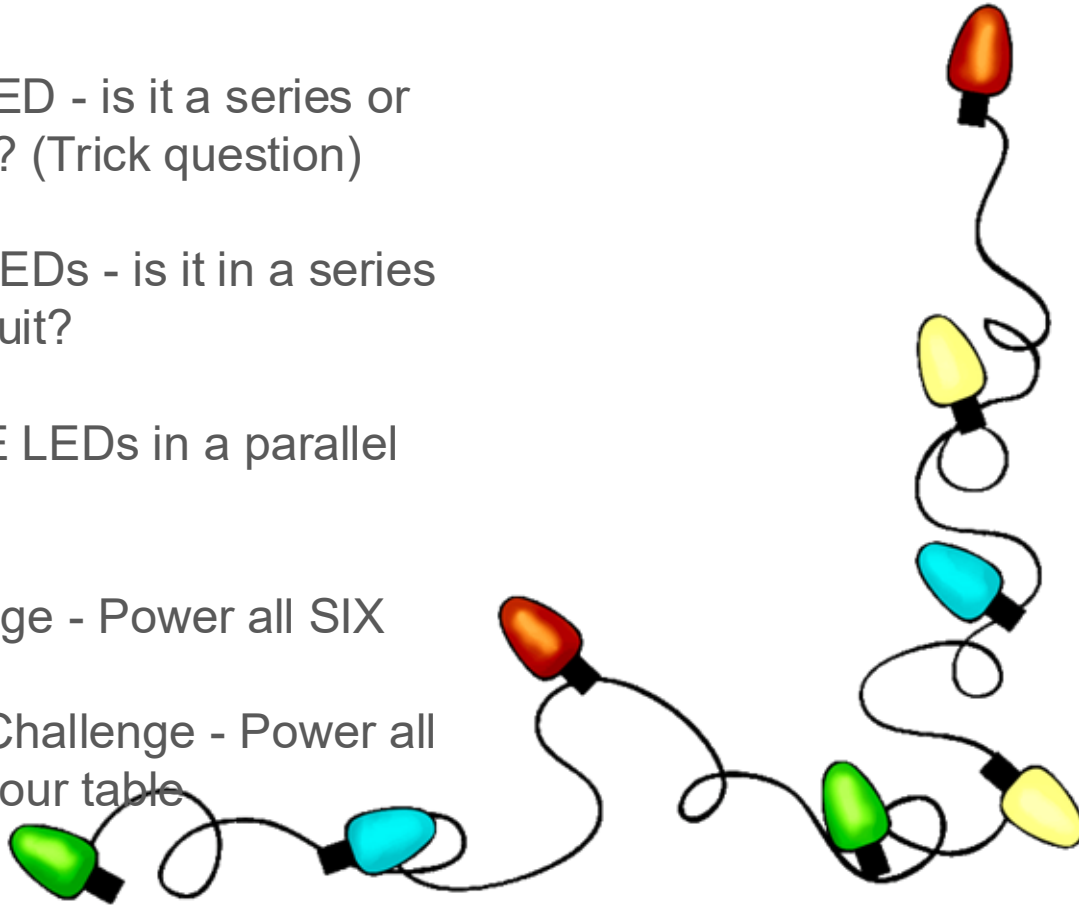
- Use the materials in the caddy to make series and parallel circuits.

Constraints:

- Leave it better than you found it for the next class.
- Time: 10 - 15 mins

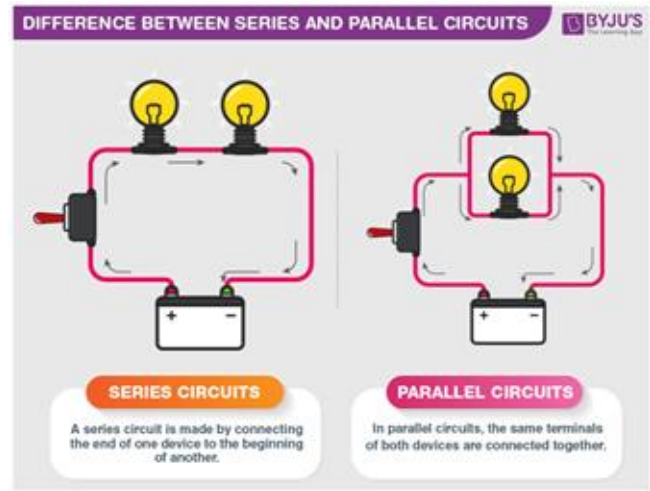
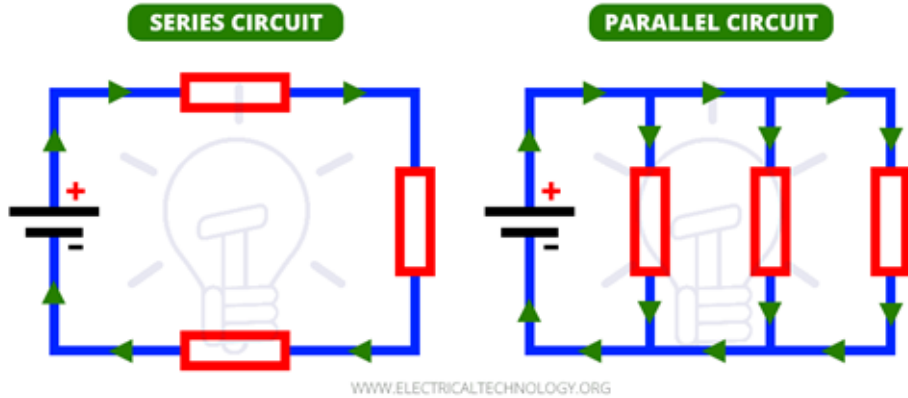
Challenge:

1. Power ONE LED - is it a series or parallel circuit? (Trick question)
1. Power TWO LEDs - is it in a series or parallel circuit?
1. Power THREE LEDs in a parallel circuit.
1. Super Challenge - Power all SIX LEDs
2. Super Super Challenge - Power all the LEDs on your table



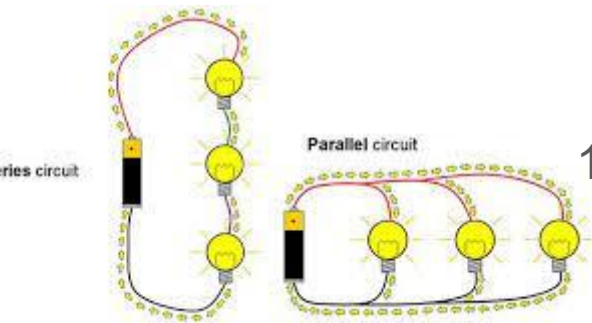
6. Super Duper Amazing Challenge - Power ALL the LEDs from ALL the tables

DIFFERENCE BETWEEN SERIES & PARALLEL CIRCUIT

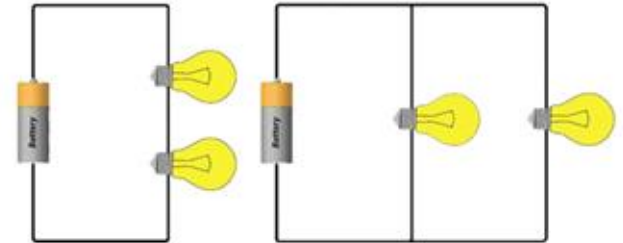


Challenge:

1. Light two LEDs with four alligator clip wires (is it series or parallel?)

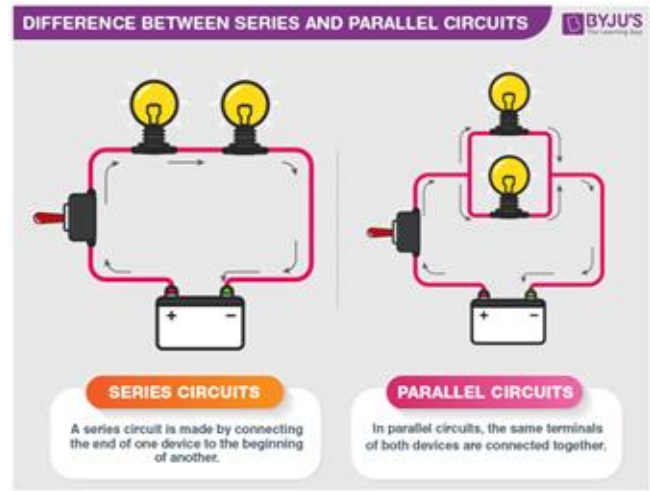
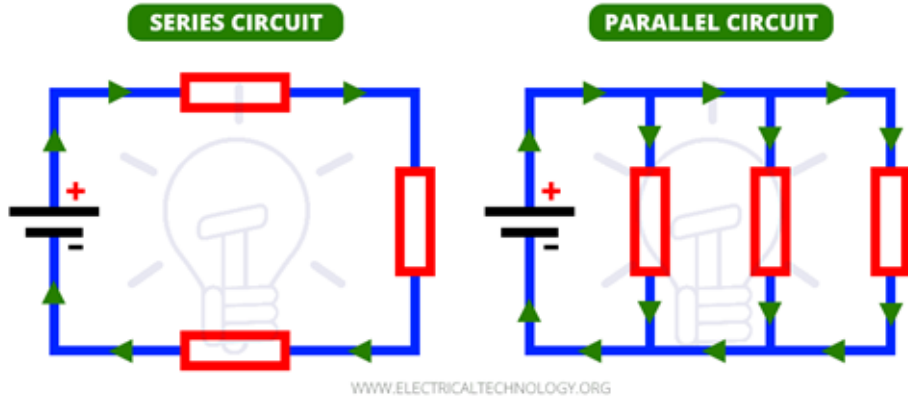


1. Light three LEDs



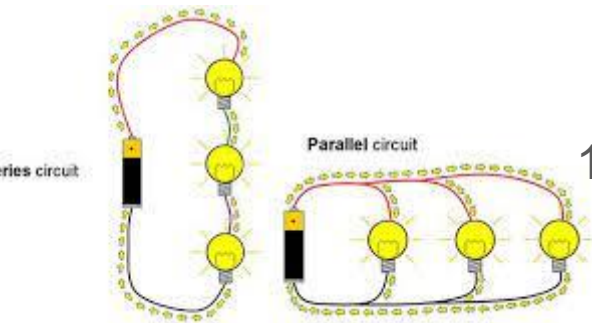
Comparison of Series and Parallel

DIFFERENCE BETWEEN SERIES & PARALLEL CIRCUIT

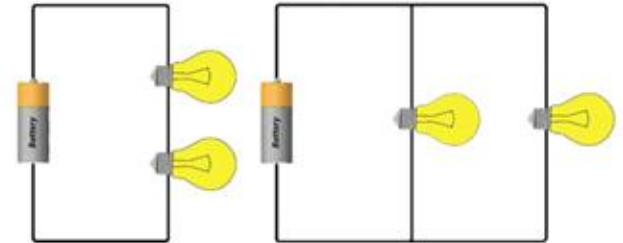


Challenge:

1. Light two LEDs with four alligator clip wires (is it series or parallel?)



1. Light three LEDs



Comparison of Series and Parallel

Name _____ Date _____

WHY WON'T IT LIGHT?

Directions: In the boxes below, tell why each light bulb will not light.



Makecode.microbit.org

Bookmark it

What is one thing you hope will happen this week?

Hexagon #33

Your goal: Make a game for your pet, friend, or family member.

Carrie goal: See how you manage time.

Requirements:

- Make a two player game to play with your client.
- It must be fun for your client.
- Must be easy to play
- We will share our makes in a gallery walk even if it's not finished.

Constraints:

- Your idea and materials list must be approved by Carrie
- You can only use the materials and tools as instructed
- Time: 15 - 20 mins

Hexagon #33

Your goal: Make a game for your pet, friend, or family member.

Carrie goal: See how you manage time.

Requirements:

- Make a two player game to play with your client.
- It must be fun for your client.
- Must be easy to play
- We will share our makes in a gallery walk even if it's not finished.

Constraints:

- Your idea and materials list must be approved by Carrie
- You can only use the materials and tools as instructed
- Time: 15 - 20 mins

What makes a good game?

Goal: Make a game that can be easily played by the whole school on the display board.

Requirements:

- Instructions must be easy
- The game must have a goal
- Game must be sturdy enough for Kindergarten play

Constraints:

- You can only use the materials given - cardboard, microbit, LEDs, wire
- 30 mins to build

Client:

Title of game:

How does a player “win”?

What are the instructions for play?

- 1.
- 2.
- 3.
- 4.
- 5.

Client:

Title of game:

How does a player “win”?

What are the instructions for play?

- 1.
- 2.
- 3.
- 4.
- 5.

Client:

Title of game:

How does a player “win”?

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Title of game:

How does a player “win”?

What are the instructions for play?

- 1.
- 2.
- 3.
- 4.
- 5.

[illegible][illegible][illegible][illegible]

Skill

Design Skills: Learning Station

Activity

What is Design? Use vs. Looks

Goal: Know the difference between functional and aesthetic design by designing a nice home for your client

Words to know:

- Functional - how something is used
- Aesthetic - how something looks and makes you feel

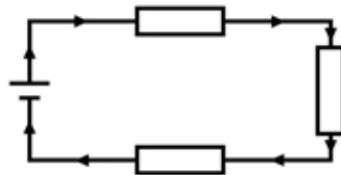
Materials:

- Clients to design for
- Pencil and color pencils
- (Optional) Device for research
- (Optional) Dotted or grid paper

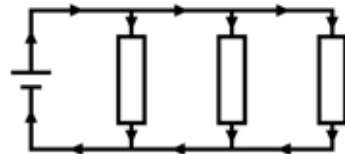
Do:

1. Pick one of the clients at the Learning Station.
2. Follow the guide for your design.
3. Reflect.

Series Circuit



Parallel Circuit



#19

Skill

Design Skills: Learning Station



Design Guide

Design Guide: What is Design? Use vs. Looks

Client Name:

Client Needs:

Client Wants (Likes):

Sketch a design with explanations labels: (Need more paper? Get it!)

HEXAGON #6



Skill

Hand Skills: Do It Together

Activity

Cardboard Cutter Safety

Goal: Learn how to use the cardboard cutter safely

Materials:

- Cardboard cutter
- Cardboard
- Surface protection (put a piece of cardboard on cutting surface)
- Optional: Pencil and ruler for planning

Do:

1. Understand the cardboard and the cardboard cutter:
 - a. Where it is sharp
 - b. How it can cut
2. Safety is paramount:
 - a. Safety for your teammates
 - b. Safety for yourself
 - c. Safety for our surfaces
3. How to hold: the cutter and the cardboard - THUMBS!
4. How to cut:
 - a. It should feel smooth
 - b. If you have to force it, you are doing it wrong - NOT SAFE!
 - i. Re-examine the cardboard
 - ii. Review how you hold the cardboard
 - iii. Cut it another way
5. Leave the work area better than you found it. If there is a name on the waiting list, please let them know when you are done.

REFLECTION

Hexagon #

Date

I learned...

I want to know more about...

Notes / Feedback



HEXAGON #7

Skill

Hand Skills: Learning Station

Activity

Use a cardboard cutter

Goal: Get a feel for using a cardboard cutter safely

Materials:

- Cardboard cutter
- Cardboard
- Surface protection (put a piece of cardboard on cutting surface)
- Optional: Pencil and ruler for planning

Do:

1. Review safety by saying it out loud to yourself:
 - a. Elbows length from teammates
 - b. Tuck thumbs
 - c. Plan carving route
 - d. When not in use, place cardboard cutter back in holder
2. Cut a square - for practicing straight cuts.
3. Cut a circle - for practicing curvy cuts.
4. Cut a star - for practicing tight cuts.
5. Leave the workspace better than you found it (usable materials back in pile scraps in trash, tools back in place, etc).

REFLECTION

Hexagon #

Date

I learned...

I want to know more about...

Notes / Feedback



Hand Skills!



Let the next person on the waiting list know it's their turn

Clean your area

Safety



How to Hold

Tight Cuts



Cardboard Cutter

Plan Cut



Curve Cut



Straight Cut



Hot Glue Gun

<https://youtu.be/UofY6lOQ-q0?si=bm79DtNtX9RfCl8i>

Paper Tower

https://www.sciencebuddies.org/science-fair-projects/project-ideas/CE_p027/civil-engineering/tallest-paper-tower-challenge

1)Straight cuts



1)Curved cuts



1)Tight cuts





CARDBOARD ATTACHMENTS



FLANGE

L-BRACE



SLOT



+



=



INSERT

TABS



+



=



BRASS FASTENERS



HEY-IT MOVES!

HOLE PUNCH
+ WIRE



FOOT

ATTACHMENT Techniques



Flange



Gusset $\Rightarrow$ Δ supports



L-brace

Slot



cut

+



cut

=



insert

Tabs



tab

+



slot



Brass Fastener



HEY-IT MOVES!

Hole Punch
+ Paperclips



skewer
(cardboard moves)

Simple Circuits

Goal: Turn on the LED

- 1) Take a template (instructions on the back)
- 2) Gather materials - please only take what you need
- 3) Make the circuit!
- 4) Put the battery back when done, clean your workspace

**** Caution: the copper tape edges can be sharp ****

AbleGamerz

PROJECT DESCRIPTION

6th grade students set out to explore the questions surrounding disability, using video gaming as both a point of common interest and a real-world engineering and technological challenge.

INSTRUCTORS

Corey Clark, Curtis Taylor

SCHOOL SITE

High Tech Middle North County

LEVEL

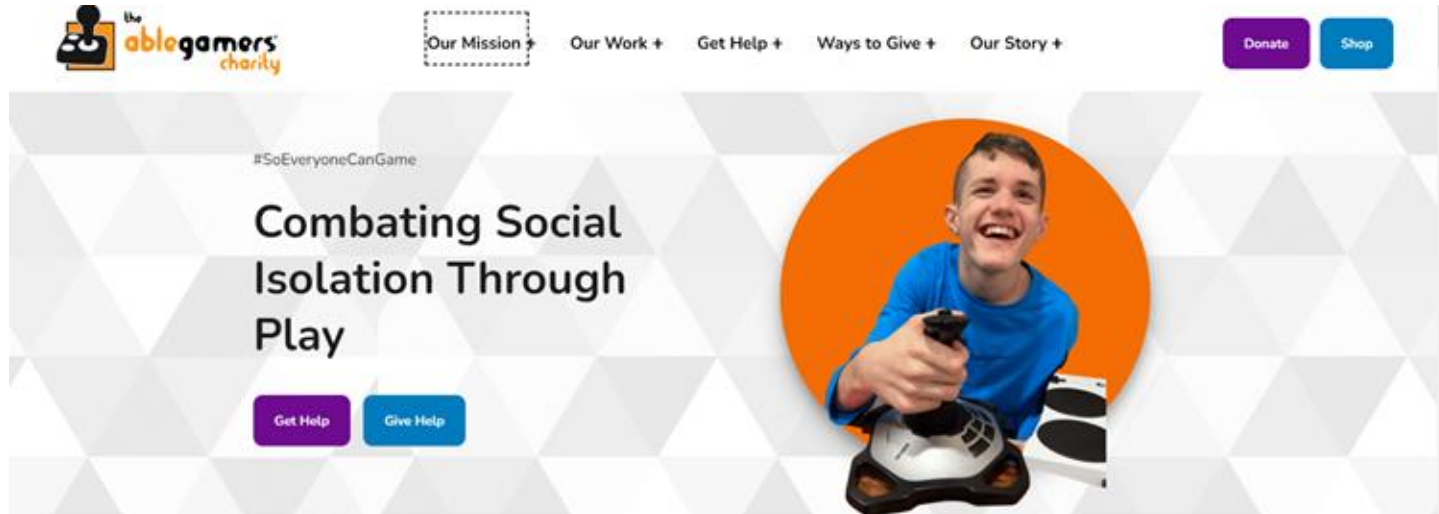
Middle School (6-8)

SUBJECTS

Arts, English, Science, Social Studies



Students were introduced to the AbleGamers organization, and went through several simulations to build empathy — being challenged to do certain tasks without the use of arms, or reading all directions through a mirror. They opened and examined some adaptive gaming controllers, reflecting on both the packaging and the adaptations. In the afternoon, the group visited a local arcade and interviewed the owner about his love of gaming and his thoughts about accessibility. Students spent the afternoon playing at the arcade, while also thinking about the accessibility issues the morning's activities had raised.



Through collaborations with community organizations working with disabled people, students were able to become familiar with particular clients. They learned about their disability and their passion for gaming and online community, and engineered a prototype for modifying a traditional gaming controller or device in a fashion that allowed greater access and ease of use.



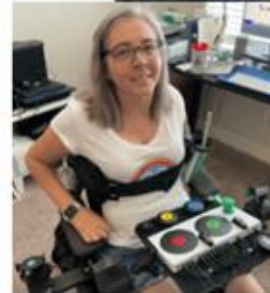
CHANGING LIVES

Creating opportunities that enable play in order to combat social isolation, foster inclusive communities, and improve the quality of life for people with disabilities.

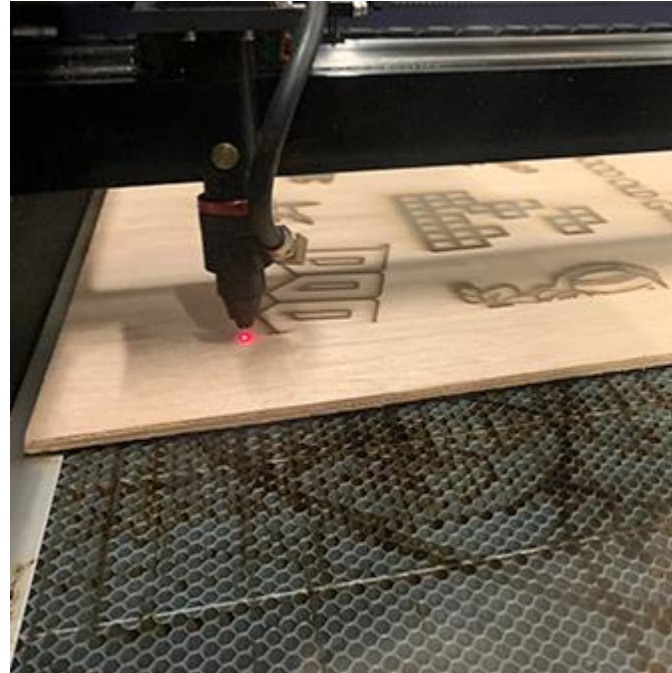
People with disabilities are 51% more likely to be socially isolated than their non-disabled peers.

You want people with disabilities to have rich well rounded lives that let them be their best selves and to have something to look forward to. Your support works every day to bring play to those desperately needing it, and we do it at no cost to the people we serve.

You Can Change a Life



Using Xbox adaptive controllers and Arduino boards connected to gaming PCs and laptops, students engineered and built real working controller devices that will allow their clients to access and play games with their tongues, feet, arms — whatever is needed to adapt to their disability (Muscular Dystrophy, amputated limbs, paralysis, etc).







FABRIKARIUM*TYO

主催・共催：



ICT Rehabilitation
Research Lab



後援：



日本科学未来館

Every year the makerspace hosts a hackathon to share, design, and make devices to help people. Makers from all over the world come to make for good.



FabLab Shinagawa is a makerspace in Japan. They specialize in designing and 3D printing devices for people with disabilities. They share their designs free on the internet for anyone to download.



Google 検索



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What would you do if you had a hard time gripping things? This 3D printed juice box holder was designed by a maker who had a grandparent who had a hard time closing their hands.



Google 検索



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パックジュースホルダー PaperPackHolder

EN | JP

IMAGES



ITEM INFORMATION

Size : 56-40-177(加工前)

Time : 2h20m

Material : PLA

DESIGN

SONOKO

LICENSE



PaperPackHolder
by SONOKO
is licensed under the BY-NC-SA
license.

FILES TO DOWNLOAD

[DOWNLOAD ALL FILES](#)



What if you couldn't hold your fork at lunch or a cup at dinner? Here are some clever designs to help the user from makers all over the world.



Fixed drink holder at any angle
どこでもドリンクホルダー

[データ | 食事]

[English](#) | [日本語](#)



T-Shape Spoon Holder L
カトラリーホルダー T型L

[データ | 食事]

[English](#) | [日本語](#)



T-Shape Spoon Holder S
カトラリーホルダー T型S

[データ | 食事]

[English](#) | [日本語](#)



Simple Spoon Holder L
カトラリーホルダー シンプル L

[データ | 食事]

[English](#) | [日本語](#)



Simple Spoon Holder M
カトラリーホルダー シンプル M

[データ | 食事]

[English](#) | [日本語](#)



Toromix
トロミックス

[データ | 食事]

[English](#) | [日本語](#)



GLASS HOLDER
グラスホルダー

[データ | 食事]

[English](#) | [日本語](#)



T-Shape Spoon Holder
カトラリーホルダー T型

[データ | 食事]

[English](#) | [日本語](#)

What if your best friend couldn't hold their soda can? What would you design and make to help them?



CanCoffeeHolder
缶ホルダー 250

[データ | 食事]

[English](#) | [日本語](#)



PaperPackHolder
パックジュースホルダー

[データ | 食事]

[English](#) | [日本語](#)



Can Holder
缶ビールホルダーA

[データ | 食事]

[English](#) | [日本語](#)



Remix Sleeve
紙コップホルダー

[データ | 食事]

[English](#) | [日本語](#)



Canned beer Holder
缶ビールホルダー B

[データ | 食事]

[English](#) | [日本語](#)



eco sleeve
エコスリーブ

[データ | 食事]

[English](#) | [日本語](#)



PetOpenerL
ペットボトルオープナー L型

[データ | 食事]

[English](#) | [日本語](#)

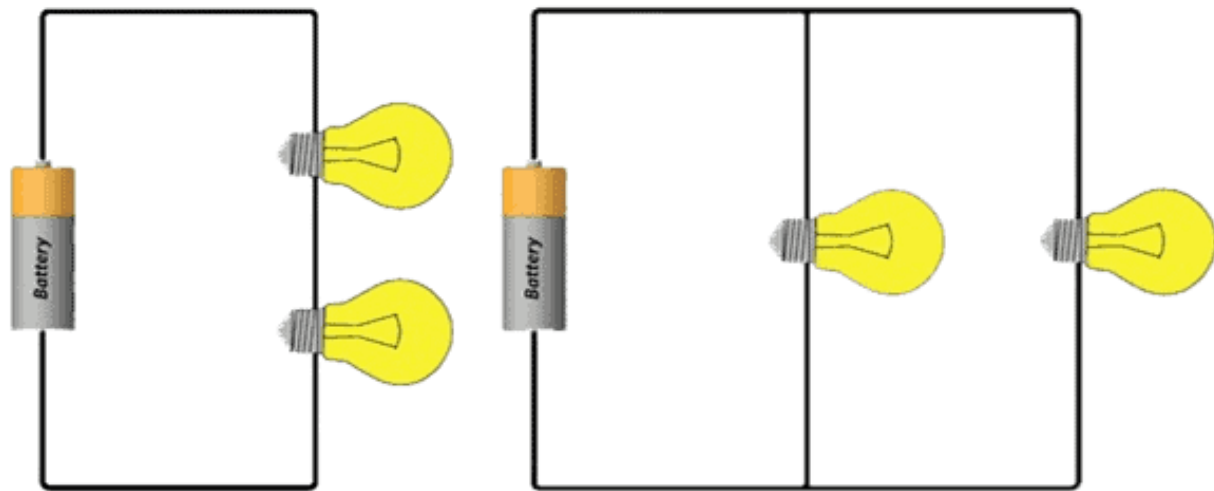


Spoon Holder with Cuff
カトラリーホルダー カフ付

[データ | 食事]

[English](#) | [日本語](#)



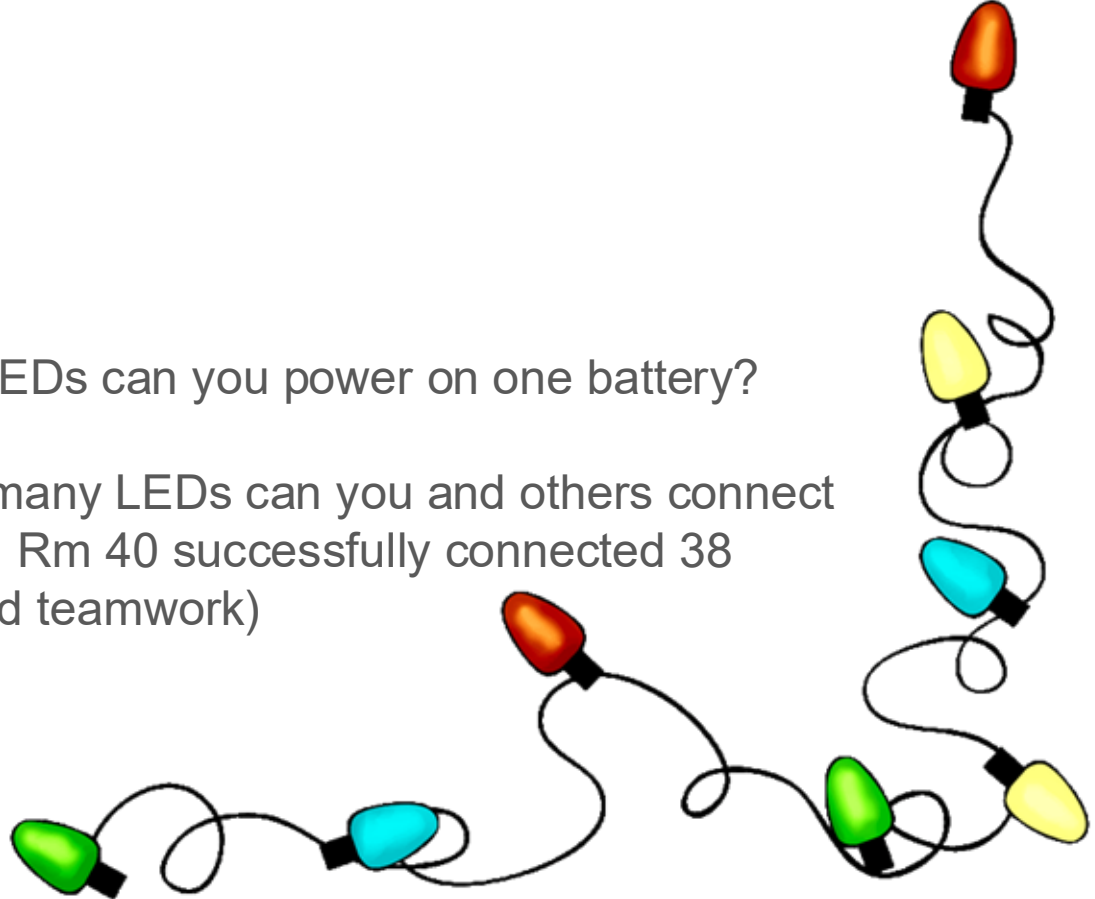


Comparison of Series and Parallel

Circuit Challenge! Can you figure out how to?

1. Power ONE LED
1. Power TWO LEDs
1. Power THREE LEDs
1. Super Challenge - How many LEDs can you power on one battery?
1. Super Duper Challenge - How many LEDs can you and others connect together? (Fun fact: 3rd Grade, Rm 40 successfully connected 38 LEDs on 4 batteries with grit and teamwork)

Helpful hints: Think parallel circuits!
Look at the poster or research it!



What do you want to see in our community?

Use cardboard engineering to make a model of the future we aspire to!

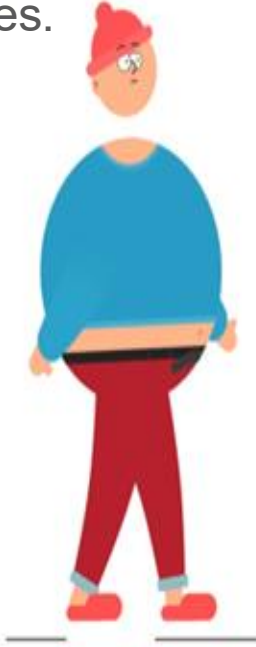
1. Make a building, park, or community area you would like to have in the neighborhood. (Use the plastic knives or scissors for cutting. Look at the poster for attachment method!)
1. Decorate it! Make sure you have a description of your building!
1. Place your building in our future community. (Inside area of the blue tape)

** Station models by the 3rd -5th grade makers.



OctoStudio Animation Week

Describe the differences in how these two characters walk. Tell one or two classmates.



Single Screen

Bird's Eye View





Single Screen

Sideview

Side Scroller

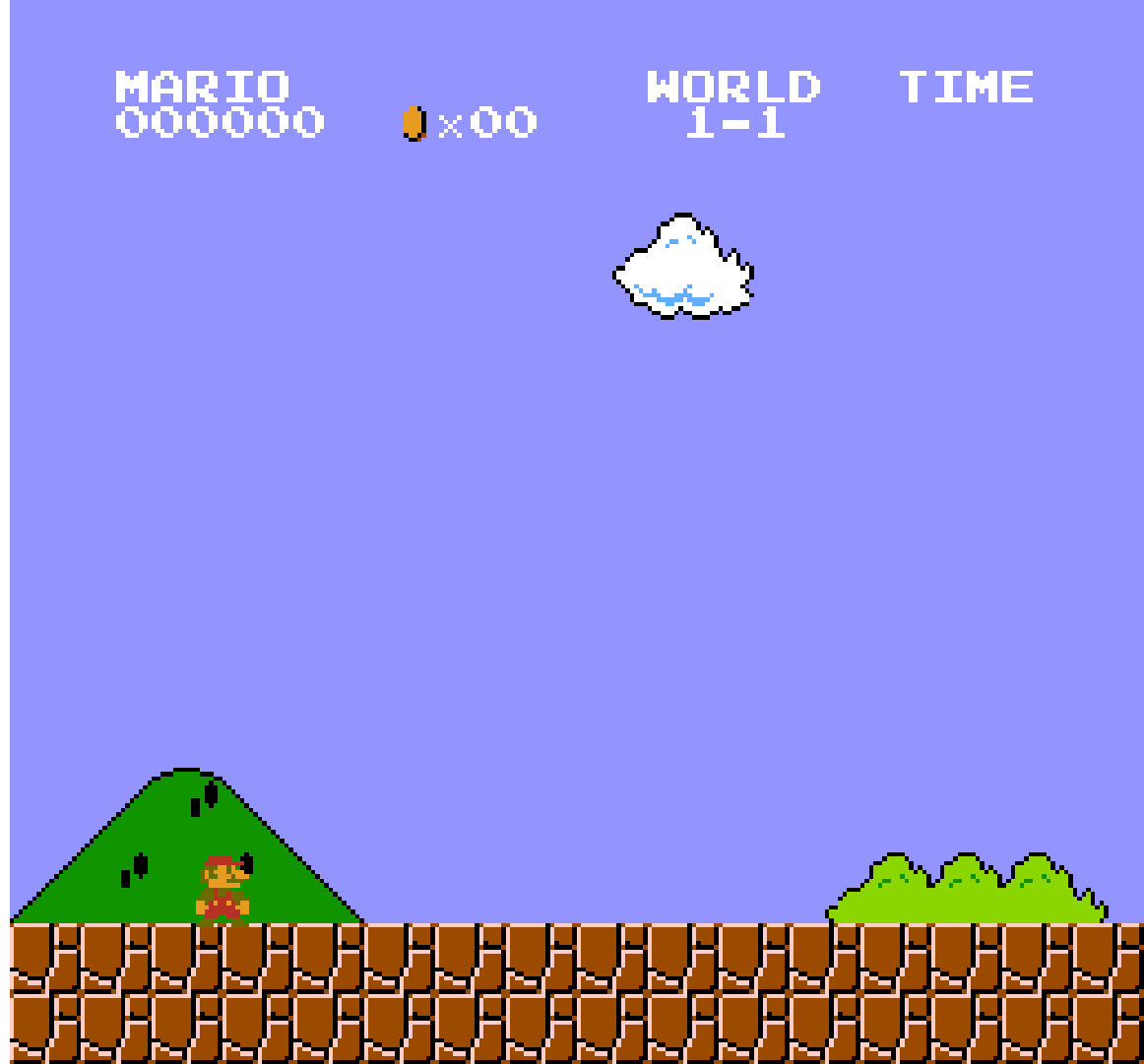


Single Screen

Sideview



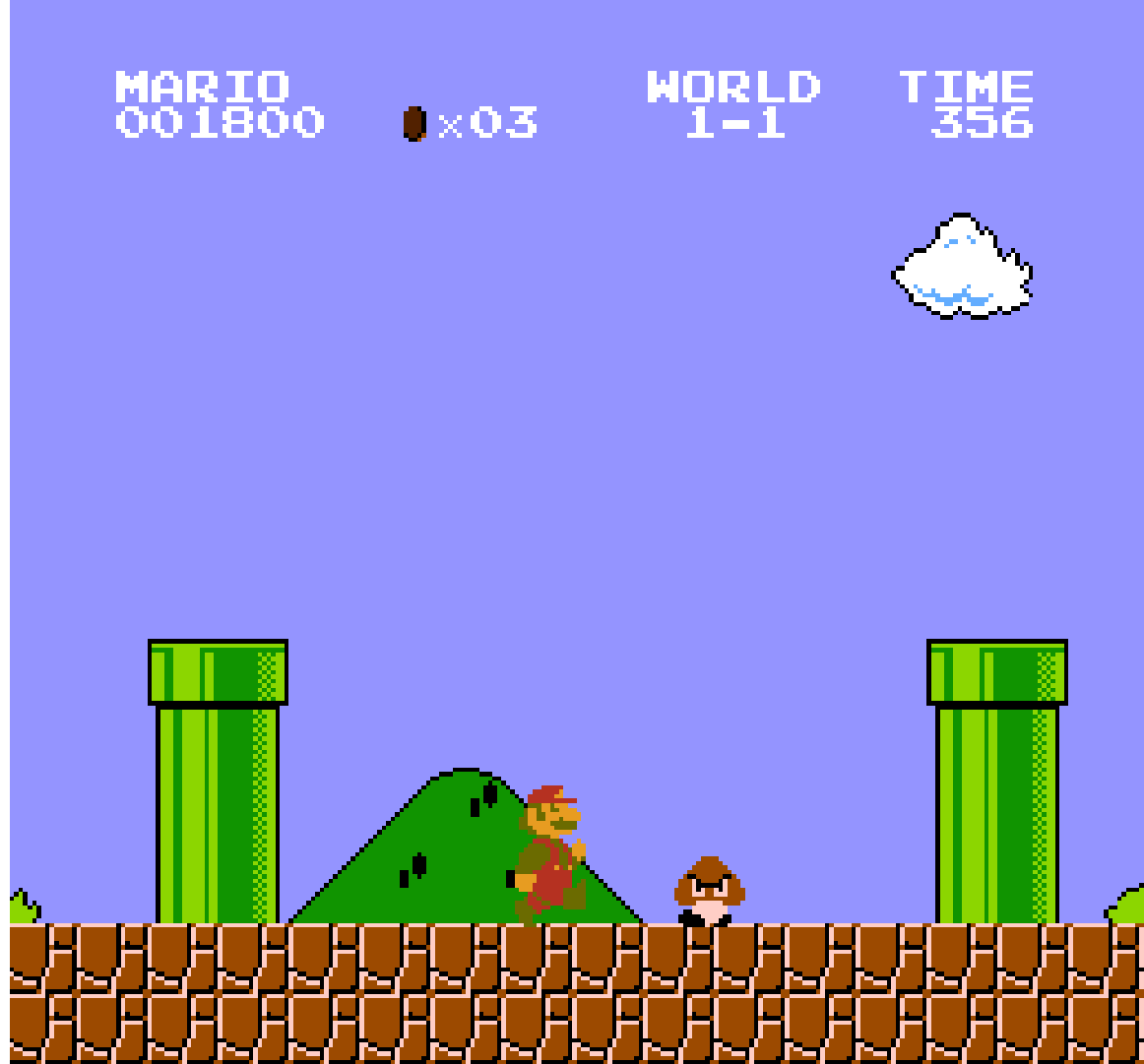
2D Platformer



2D Platformer

With

Single Screen
Platformer “Level”



Week 3

Video game movement discussion

OctoStudio Demonstration

iPad rules

Challenge 1 - Make a character jump in a loop. Done? What else can you add?

Challenge 2 - Design your own character. Demo of hand drawn.